CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International General Certificate of Secondary Education

MARK SCHEME for the October/November 2015 series

0625 PHYSICS

0625/22 Paper 2 (Core Theory), maximum raw mark 80

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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NOTES ABOUT MARK SCHEME SYMBOLS AND OTHER MATTERS

B marks — are independent marks, which do not depend on any other marks. For a B mark to be scored, the point to which it refers must actually be seen in the candidate's answer.

M marks — are method marks upon which accuracy marks (A marks) later depend. For an M mark to be scored, the point to which it refers **must** be seen in a candidate's answer. If a candidate fails to score a particular M mark, then none of the dependent A marks can be scored.

C marks — are compensatory method marks which can be scored even if the points to which they refer are not written down by the candidate, provided subsequent working gives evidence that they must have known it. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows he or she knew the equation, then the C mark is scored.

A marks — are accuracy or answer marks which either depend on an M mark, or which are one of the ways which allow a C mark to be scored.

Brackets ( ) — around words or units in the mark scheme are intended to indicate wording used to clarify the mark scheme, but the marks do not depend on seeing the words or units in brackets, e.g. 10 (J) means that the mark is scored for 10, regardless of the unit given.

c.a.o. — means "correct answer only".

e.c.f. — means "error carried forward". This indicates that if a candidate has made an earlier mistake and has carried his incorrect value forward to subsequent stages of working, he or she may be given marks indicated by e.c.f. provided his or her subsequent working is correct, bearing in mind his or her earlier mistake. This prevents a candidate being penalised more than once for a particular mistake, but **only** applies to marks annotated "e.c.f."

e.e.o.o. — means "each error or omission".

owtte — means “or words to that effect”

<u>Underlining</u> — indicates that this <u>must</u> be seen in the answer offered, or something very similar.

OR — indicates alternative answers, any one of which is satisfactory for scoring the mark.

AND — indicates that both answers are required to score the mark.

Spelling — Be generous about spelling and use of English. However, do not allow ambiguities, e.g. spelling which suggests confusion between reflection / refraction / diffraction or thermistor / transistor / transformer.

Significant figures — Answers are generally acceptable to any number of significant figures ≥ 2, except where the mark scheme specifies otherwise.

Units — On this paper, incorrect units are not penalised, except where specified. More commonly, marks are allocated for specific units.

Fractions — These are only acceptable where specified.

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NOT indicates that an incorrect answer is not to be disregarded, but cancels another otherwise correct alternative offered by the candidate. i.e. right plus wrong penalty applies.

ROOT 19

1	**(a)**	any **two** from: wire not starting at 0 cm wire not straight wire away from / not close to rule	B2
	(b) (i)	0.65 (cm)	B1
	(ii)	candidate's **(b)(i)** / 8 0.8125 OR 0.813 OR e.c.f. 0.81	C1 C1 A1
			[Total: 6]
2	**(a) (i)**	10 (m / s)	B1
	(ii)	distance = speed × time OR 10 × 20 200 (m)	C1 A1
	(iii)	7 (s)	B1
	(b) (i)	50 (N) forwards	B1 B1
	(ii)	(cyclist is) accelerating / speed increases	B1
	(iii)	move more slowly / speed decreases / decelerates	B1
			[Total: 8]
3	**(a)**	decreases	B1
	(b)	increases	B1
	(c)	increases	B1
	(d)	increases does not change decreases	B1 B1 B1
			[Total: 6]

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4	**(a)**	**(i)**	joule / J / kJ	B1
		(ii)	kinetic	B1
		(iii)	heat / thermal sound	B2
		(iv)	more efficient.	B1
	(b)		greater area (of tyres in contact with ground)	B1
			less pressure	B1
			weight / force spread out OR $p = F / A$	B1

[Total: 8]

5	**(a)**		strip bent upwards	B1
	(b)	**(i)**	two fixed points marked	B1
			ice or steam point stated	B1
			positions divided by difference in temperature OR the more it bends the higher the temperature	B1
		(ii)	plausible suggestion, e.g. inaccurate, too large, difficult to calibrate	B1

[Total: 5]

6	**(a)**	**(i)**	at least two arrows pointing in correct direction	B1
		(ii)	convection	B1
	(b)		hot water expands / water molecules further apart	B1
			hot water less dense	B1
			hot water rises	B1
			cool water falls / takes place of hot water	B1

[Total: 6]

7	**(a)**	**(i)**	sound travels slowly	B1
			sound travels slower than light or reverse argument	B1
		(ii)	reflection of the sound from the cliff	B1
	(b)		evidence of average found / 1.56 OR 1.6	B1
			speed = distance / time in any form: words, symbols, numbers	C1
			500 / 1.6 OR 500 / candidate's time	C1
			321 OR e.c.f.	A1

[Total: 7]

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8	**(a)**	**(i)**	F_2 correctly positioned and labelled	B1
		(ii)	ray through centre C of the lens correctly drawn	B1
		(iii)	second ray correctly drawn through either principal focus and horizontal section, correct to better than ± 1 small square	B1
		(iv)	position of inverted image shown	B1
	(b)		diminished inverted real	B3
				[Total: 7]
9	**(a)**	**(i)**	a.c. waveform drawn	B1
	(b)	**(i)**	(step-down) transformer	B1
		(ii)	$V_1 / V_2 = N_1 / N_2$ 4800 / (120 / 10) OR correct substitution 400	B1 C1 A1
				[Total: 5]
10	**(a)**	**(i)**	iron core becomes an electromagnet (bar magnet is) repelled	B1 B1
		(ii)	steel	B1
	(b)		move pivoted magnet to new position mark direction of arrow OR mark direction N pole points repeat for other positions around magnet	B1 B1 B1
				[Total: 6]

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11 **(a)** motor / blower and heater — B3
motor / blower only
nothing / none OR left blank
nothing / none OR left blank

Any three correct for 2 marks.
Any two correct for 2 marks.

(b) $V = IR$ — B1

(c) 250 / 2 — C1
125 (Ω) — A1

(d) **(i)** fuse symbol correct and placed correctly — B1

(ii) (fuse) wire melts — B1
circuit breaks / incomplete — B1

[Total: 9]

12 **(a)** nature: helium nucleus owttte, (e.g. 2p + 2n) — B1
charge: minus 1 / –1 / 1– / negative — B1

(b) **(i)** (α particles) produce more ions (/ cm) — B1
(α particles) collide with / are stopped by smoke — B1

(ii) 100 years — B1

(c) 237 — B1
93 — B1

[Total: 7]

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